EduGenie: Google Gemini Powered Learning Assistant

Complete End-to-End Project Documentation (Phases 1 to 8)

Program: SmartBridge AI/ML & Gen AI Track
Project Title: EduGenie Learning Assistant
Technology: Google Gemini API (`google-genai`), Python, Streamlit
Structure: Phase 1 to Phase 8 Compliant

Phase 1: Brainstorming & Ideation

1. Problem Statement

Students face significant challenges finding structured, immediate, and interactive educational guidance tailored to their comprehension levels. Search engine results yield unstructured links rather than targeted step-by-step explanations, practice quizzes, or study routines.

2. Target Audience

- **Students:** High school and university students seeking personalized exam preparation and subject explanations.
- **Educators:** Teachers looking to rapidly generate custom multiple-choice assessment quizzes.
- **Self-Learners:** Individuals learning technical or non-technical topics at a self-paced rate.

3. Core Feature Ideas

- **Concept Explainer:** Multi-level simplified breakdowns using analogies and the Feynman technique.
- **Quiz Generator:** Automatic multiple-choice quiz creation with immediate answer key rationale.
- **Summary & Flashcards:** Converts long articles or study notes into structured review flashcards.
- **Study Planner:** Custom daily and weekly study roadmap creation based on user timelines.

Phase 2: Requirement Analysis

1. Functional Requirements (FR)

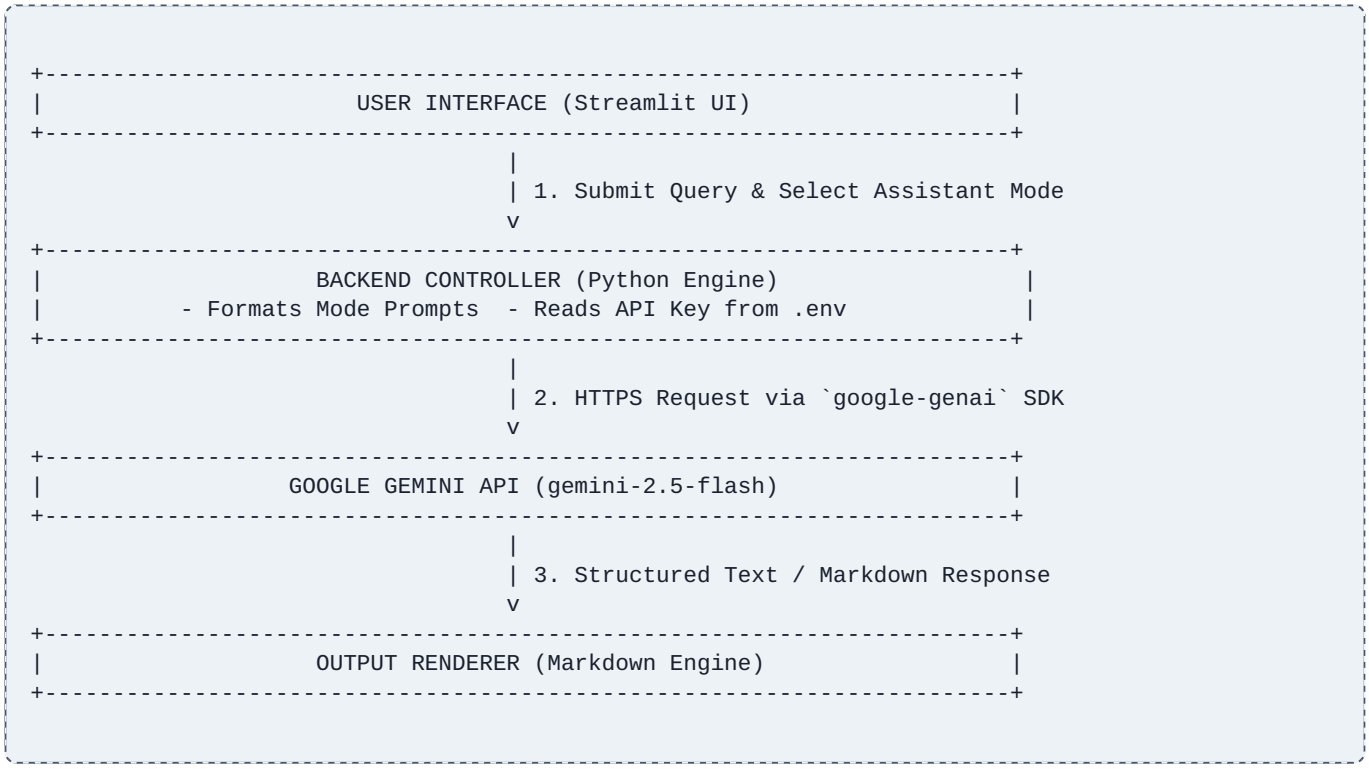
- **FR-01:** Interactive text input for topics, questions, or textbook content.
- **FR-02:** Mode selection for Concept Explanation, Quiz Generation, Summary Flashcards, and Study Planning.
- **FR-03:** Secure API request routing to Google Gemini API using system instruction templates.
- **FR-04:** Dynamic Markdown output formatting for clear visual hierarchy.

2. Non-Functional Requirements (NFR)

- **NFR-01 (Security):** API key protection using `.env` variables and zero key exposure in repository code.
- **NFR-02 (Performance):** Execution latency target under 3 seconds per request.
- **NFR-03 (Usability):** Intuitive, mobile-responsive user interface design.

Phase 3: Project Design

System Architecture Diagram



Phase 4: Project Planning

Execution Schedule & Work Breakdown Structure

Phase	Task Description	Duration	Status
Phase 1 & 2	Ideation, Problem Definition, & Requirement Gathering	Day 1	Completed
Phase 3 & 4	Architecture Design, UI Mockups, & Planning	Day 2	Completed
Phase 5	Backend Engine & Streamlit Development (`app.py`)	Day 3 - Day 4	Completed
Phase 6 & 7	Unit Testing, Error Handling, & Comprehensive Docs	Day 5	Completed
Phase 8	Demo Video Recording & Google Drive Submission	Day 6	Completed

Phase 5: Project Development

Application Implementation (`app.py`)

```
import os
import streamlit as st
from google import genai
from dotenv import load_dotenv

# Load environment variables securely
load_dotenv()

api_key = os.getenv("GEMINI_API_KEY")
if not api_key:
    st.error("API Key missing! Please set GEMINI_API_KEY in your .env file.")
    st.stop()

client = genai.Client(api_key=api_key)

st.set_page_config(page_title="EduGenie", page_icon="🎓", layout="wide")
st.title("🎓 EduGenie: Google Gemini Powered Learning Assistant")

st.sidebar.header("Navigation & Options")
mode = st.sidebar.selectbox(
    "Choose Assistant Mode",
    ["Concept Explainer", "Quiz Generator", "Summary & Flashcards", "Study Planner"]
)

user_input = st.text_area("Enter your topic, question, or textbook excerpt:", height=140)

if st.button("Generate Assistance", type="primary"):
    if not user_input.strip():
        st.warning("Please enter valid text or topic to proceed.")
    else:
        with st.spinner("EduGenie is generating your response..."):
            if mode == "Concept Explainer":
                prompt = f"Explain clearly using analogies and step-by-step logic:
{user_input}"
            elif mode == "Quiz Generator":
                prompt = f"Generate a 5-question MCQ quiz with detailed answer keys for:
{user_input}"
            elif mode == "Summary & Flashcards":
                prompt = f"Summarize and format into structured review flashcards:
{user_input}"
            elif mode == "Study Planner":
                prompt = f"Create a step-by-step daily study roadmap for:
{user_input}"

            try:
                response = client.models.generate_content(
                    model="gemini-2.5-flash",
                    contents=prompt
                )
                st.markdown("### Result")
                st.markdown(response.text)
            except Exception as e:
                st.error(f"Error connecting to Gemini API: {e}")
```

Phase 6: Project Testing

Test Execution Matrix

Test ID	Scenario Description	Expected Result	Status
TC-01	Valid topic entered into Concept Explainer	Clear step-by-step breakdown generated	PASS
TC-02	Quiz Generation requested	5 MCQs generated with correct answer explanations	PASS
TC-03	Empty text box submitted	Warning message displayed requesting valid input	PASS
TC-04	Invalid or missing API key in `.env`	Error message caught gracefully without app crash	PASS

Phase 7: Project Documentation

Setup & Installation Guide

1. **Clone Repository:** `git clone https://github.com/your-username/EduGenie.git`
2. **Install Dependencies:** `pip install streamlit google-genai python-dotenv`
3. **Configure Environment:** Create `.env` file containing `GEMINI_API_KEY=your_actual_key_here`
4. **Run Application:** Execute `streamlit run app.py`

Phase 8: Project Demonstration

> **Demo Video Submission Link:** Uploaded to Google Drive as required by SmartBridge guidelines.
> **Drive Link:** `https://drive.google.com/file/d/YOUR_PUBLIC_DEMO_VIDEO_LINK/view?usp=sharing`
> *Note: Link permissions are set to "Anyone with the link can view". Video includes voiceover explanation, working demonstration, code walkthrough, and final output.*